

Document ID	Revision	Applies To	Status
FPT-2026-0114	v2.3.0	SY32L-S1, Rev. C silicon	Released

Scope and Audience

This document is the firmware-facing register and interrupt reference for the Synapse SY32L-S1 microcontroller family. It is written for embedded engineers who are bringing up board support packages, writing peripheral drivers, or debugging interrupt latency on Rev. C silicon. It does not cover electrical characteristics, package mechanicals, or qualification data – those live in the SY32L-S1 Datasheet (FPT-2026-0110). Register bit layouts, reset values, and access types below are normative for firmware and take precedence over any inline code comments in the reference driver package.

Readers are assumed to be comfortable with ARM Cortex-M0+ programming, memory-mapped I/O, and nested vectored interrupt controllers. Address values are given in hexadecimal without separators (e.g. 0x08000000). Bit ranges are written [MSB:LSB]; a single-bit field is written with one index only.

Register Notation

RW read/write **RO** read-only **RC** read-clears (write ignored) **W1C** write-1-to-clear **Res.** reserved, must be written as reset value. Undefined behaviour results from writing reserved bits to anything other than their reset value.

1 Device Architecture

The SY32L-S1 pairs a Cortex-M0+ core running at up to 64 MHz with a single-cycle AHB matrix and two APB peripheral bridges. Firmware talks to every peripheral in this document through memory-mapped registers on APB1 or APB2; there are no port-mapped I/O instructions on this core.